

Exercise Set: Simple Quantum Circuits with Qiskit

Instructions

This exercise set combines:

- manual calculations of quantum states and gate operations,
- understanding and prediction of Qiskit programs,
- implementation of quantum gates and circuits using Qiskit,
- comparison of theoretical and simulated results.

Students should perform manual derivations before running the corresponding Qiskit programs wherever mentioned.

1 Qubits and State Vectors

Exercise 1: Basis states

Write the vector representation of the following states manually:

1. $|0\rangle$
2. $|1\rangle$
3. $|00\rangle$
4. $|01\rangle$
5. $|10\rangle$
6. $|11\rangle$

Using the tensor product, prove that

$$|01\rangle = |0\rangle \otimes |1\rangle = \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix}.$$

Exercise 2: Normalization of a qubit

Determine whether each state is a valid normalized qubit state.

(a)

$$|\psi\rangle = \frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle$$

(b)

$$|\psi\rangle = \frac{1}{2}|0\rangle + \frac{\sqrt{3}}{2}|1\rangle$$

(c)

$$|\psi\rangle = \frac{1}{2}|0\rangle + \frac{1}{2}|1\rangle$$

Use

$$|c_0|^2 + |c_1|^2 = 1$$

to justify your answer.

Then write Python code that verifies normalization numerically.

Exercise 3: Construct a state from θ and ϕ

Given

$$|\psi\rangle = \cos\left(\frac{\theta}{2}\right)|0\rangle + e^{i\phi}\sin\left(\frac{\theta}{2}\right)|1\rangle,$$

calculate the state manually for:

1. $\theta = 0, \phi = 0$
2. $\theta = \pi, \phi = 0$
3. $\theta = \frac{\pi}{2}, \phi = 0$
4. $\theta = \frac{\pi}{2}, \phi = \pi$

Then implement:

```
def state(theta, phi):  
    ...
```

The function should return the two amplitudes.

Exercise 4: Measurement probabilities

For

$$|\psi\rangle = \frac{\sqrt{3}}{2}|0\rangle + \frac{1}{2}|1\rangle,$$

calculate manually:

$$P(0) \quad \text{and} \quad P(1).$$

Then prepare the state in Qiskit and simulate 1000 measurements. Compare the experimental frequencies with the theoretical probabilities.

2 Single-Qubit Gates

Exercise 5: Pauli-X gate

Given

$$X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix},$$

calculate manually:

$$X|0\rangle \quad \text{and} \quad X|1\rangle.$$

Then implement both cases using Qiskit.

Complete:

```
from qiskit import QuantumCircuit
from qiskit.quantum_info import Statevector

qc = QuantumCircuit(1)

# Add the required gate

state = Statevector.from_instruction(qc)
print(state)
```

Exercise 6: Two consecutive X gates

Without running Qiskit first, predict the final state of

$$X(X|0\rangle).$$

Then verify using:

```
qc.x(0)
qc.x(0)
```

Explain why

$$X^2 = I.$$

Exercise 7: Hadamard gate

Given

$$H = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix},$$

calculate manually:

$$H|0\rangle \quad \text{and} \quad H|1\rangle.$$

Then reproduce both cases in Qiskit and display the statevector.

Exercise 8: Two Hadamard gates

Calculate manually:

$$H(H|0\rangle).$$

Then create a Qiskit circuit containing two consecutive Hadamard gates:

$$|0\rangle \longrightarrow H \longrightarrow H.$$

What is the final state? Show that

$$H^2 = I.$$

Exercise 9: Pauli-Z gate

Calculate:

$$Z|0\rangle \quad \text{and} \quad Z|1\rangle$$

for

$$Z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}.$$

Then determine manually what happens in the circuit

$$|0\rangle \xrightarrow{H} \xrightarrow{Z}.$$

Verify your result using Qiskit.

Exercise 10: Phase versus measurement

Consider:

$$|\psi_1\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}$$

and

$$|\psi_2\rangle = \frac{|0\rangle - |1\rangle}{\sqrt{2}}.$$

Answer:

1. What is $P(0)$ for each state?
2. What is $P(1)$ for each state?
3. Are the two states identical?
4. Why can computational-basis measurement not distinguish them directly?

Create both states using Qiskit.

3 Understanding Qiskit Code

Exercise 11: Predict the output

Study the following code without running it first:

```
qc = QuantumCircuit(1)

qc.x(0)
qc.h(0)

state = Statevector.from_instruction(qc)

print(state)
```

Answer:

1. What is the initial state?
2. What is the state after $x(0)$?
3. What is the final state after $h(0)$?
4. What statevector do you expect?

Then run the code and verify your prediction.

Exercise 12: Debug the circuit

The intention is to prepare

$$\frac{|0\rangle + |1\rangle}{\sqrt{2}}.$$

A student writes:

```
qc = QuantumCircuit(1)
qc.x(0)

state = Statevector.from_instruction(qc)
```

Identify the mistake and correct the code.

Exercise 13: Understand circuit ordering

Consider:

```
qc = QuantumCircuit(1)

qc.h(0)
qc.z(0)
qc.h(0)
```

First calculate the result manually. Then verify using Qiskit.

What familiar gate is equivalent to

$$HZH?$$

4 Multiple Qubits and Tensor Products

Exercise 14: Tensor-product calculation

Calculate manually:

$$|0\rangle \otimes |1\rangle.$$

Use

$$|0\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad |1\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix}.$$

Then verify using:

```
np.kron()
```

Exercise 15: Product of superposition states

Calculate:

$$|+\rangle \otimes |+\rangle,$$

where

$$|+\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}.$$

Express the result in the computational basis. Then implement a two-qubit Qiskit circuit that generates this state.

Exercise 16: Three-qubit superposition

Start with

$$|000\rangle.$$

Apply a Hadamard gate to every qubit.

Determine manually the final state. Then implement:

```
qc = QuantumCircuit(3)
```

How many computational-basis states have nonzero amplitude? What is the amplitude of each?

5 Controlled Gates and Entanglement

Exercise 17: CNOT truth table

Construct the complete truth table for a CNOT gate. Assume the first qubit is the control and the second is the target.

Determine:

$$|00\rangle \rightarrow?, \quad |01\rangle \rightarrow?, \quad |10\rangle \rightarrow?, \quad |11\rangle \rightarrow?.$$

Then verify each case using Qiskit.

Exercise 18: Create a Bell state

Starting from

$$|00\rangle,$$

apply:

1. H to qubit 0.
2. CNOT with qubit 0 as control and qubit 1 as target.

Calculate the state after each step. Show that the final state is

$$|\Phi^+\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}}.$$

Implement the circuit in Qiskit and display both the circuit and the statevector.

Exercise 19: Bell-state measurement

Modify the Bell-state circuit by adding measurements. Run it for:

- 10 shots,
- 100 shots,
- 1000 shots,
- 10,000 shots.

Record the number of occurrences of 00 and 11. Explain why 01 and 10 should ideally not occur.

Exercise 20: Generate another Bell state

Design a circuit that generates

$$|\Psi^+\rangle = \frac{|01\rangle + |10\rangle}{\sqrt{2}}.$$

Tasks:

1. Determine which additional X gate is required.
2. Draw the circuit manually.
3. Implement it in Qiskit.
4. Verify the statevector.
5. Measure the circuit.

6 SWAP and Toffoli Gates

Exercise 21: SWAP operation

Prepare the state

$$|01\rangle.$$

Apply a SWAP gate. Predict the resulting state.

Then implement:

```
qc.swap(0, 1)
```

Verify the result.

Exercise 22: SWAP decomposition

A SWAP gate can be constructed using three CNOT gates:

```
CNOT(q0,q1)
CNOT(q1,q0)
CNOT(q0,q1)
```

Test the circuit for:

$$|00\rangle, |01\rangle, |10\rangle, |11\rangle.$$

Compare the results with Qiskit's built-in:

```
qc.swap(0, 1)
```

Exercise 23: Toffoli gate truth table

The Toffoli gate has two control qubits and one target qubit.

Create its complete eight-row truth table for

$$|000\rangle$$

through

$$|111\rangle.$$

Identify the input combinations for which the target qubit changes.

Then verify using:

```
qc.ccx(0, 1, 2)
```

7 Measurements

Exercise 24: Hadamard measurement

Construct:

```
qc = QuantumCircuit(1, 1)

qc.h(0)
qc.measure(0, 0)
```


Before running the simulator, predict the measurement distribution.
Then execute for 1000 shots and calculate:

$$P(0) = \frac{N_0}{N}, \quad P(1) = \frac{N_1}{N}.$$

Compare the results with theory.

Exercise 25: Deterministic versus probabilistic circuits

Compare the following circuits.

Circuit A

```
qc.x(0)
qc.measure(0, 0)
```

Circuit B

```
qc.h(0)
qc.measure(0, 0)
```

For each circuit:

1. Predict the outcome.
2. Run 1000 shots.
3. Explain why one circuit is deterministic and the other probabilistic.

8 Circuit Mapping

Exercise 26: Quantum XOR circuit

Classically,

$$S = A \oplus B.$$

Design a reversible quantum circuit using three qubits:

$$|A, B, 0\rangle$$

such that the third qubit becomes

$$A \oplus B.$$

Tasks:

1. Determine the required gates.
2. Draw the circuit.
3. Implement it in Qiskit.
4. Test all four combinations of A and B .

Exercise 27: Quantum AND using Toffoli

Use three qubits:

$$|A, B, 0\rangle.$$

Design the circuit such that the third qubit becomes

$$A \wedge B.$$

Test:

A	B	Expected output
0	0	0
0	1	0
1	0	0
1	1	1

Use the Toffoli gate.

9 Quantum Half-Adder

Exercise 28: Manual half-adder design

A classical half-adder produces

$$S = A \oplus B$$

and

$$C = A \wedge B.$$

Using four qubits

$$|A, B, S, C\rangle,$$

where initially

$$S = C = 0,$$

determine which quantum gates are required to calculate the sum and carry.

Do not use Qiskit initially. Draw the circuit manually.

Exercise 29: Implement the half-adder

Implement the circuit from Exercise 28 using Qiskit.

Use:

```
QuantumCircuit(4, 4)
```

Your circuit must use:

- two CNOT gates for the sum,
- one Toffoli gate for the carry.

Test all four combinations:

$$00, \quad 01, \quad 10, \quad 11.$$

Create a table comparing theoretical and simulated results.

Exercise 30: Verify the half-adder equation

For each input combination, verify:

$$A + B = S + 2C.$$

Write Python code that automatically checks this equation for all four possible inputs. The program should print either **PASS** or **FAIL** for each case.

10 Code Completion Exercises

Exercise 31: Complete the Bell-state program

Complete the missing statements:

```
from qiskit import QuantumCircuit
from qiskit.quantum_info import Statevector

qc = QuantumCircuit(2)

# Step 1
-----

# Step 2
-----

state = Statevector.from_instruction(qc)

print(state)
display(qc.draw("mpl"))
```

The required output state is

$$\frac{|00\rangle + |11\rangle}{\sqrt{2}}.$$

Exercise 32: Complete the measurement program

Complete the code so that a qubit is prepared in an equal superposition and measured 1000 times.

```
from qiskit import QuantumCircuit
from qiskit_aer import AerSimulator

qc = QuantumCircuit(1, 1)

-----
-----

simulator = -----

result = simulator.run(
    qc,
```

```
shots=-----  
) .result()  
  
counts = -----  
  
print(counts)
```

11 Predict Before Running

For each circuit below, do not run Qiskit first. Calculate the final state manually and then verify it.

Exercise 33

```
qc = QuantumCircuit(1)  
qc.x(0)  
qc.z(0)
```

Exercise 34

```
qc = QuantumCircuit(1)  
qc.h(0)  
qc.x(0)
```

Exercise 35

```
qc = QuantumCircuit(1)  
qc.h(0)  
qc.z(0)  
qc.h(0)
```

Exercise 36

```
qc = QuantumCircuit(2)  
  
qc.x(0)  
qc.cx(0, 1)
```

Determine the final computational-basis state.

Exercise 37

```
qc = QuantumCircuit(2)

qc.h(0)
qc.cx(0, 1)
qc.x(1)
```

Find the final state algebraically and identify the corresponding Bell state.

12 Challenge Problems

Exercise 38: Create all four Bell states

Create Qiskit circuits for:

$$|\Phi^+\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}},$$

$$|\Phi^-\rangle = \frac{|00\rangle - |11\rangle}{\sqrt{2}},$$

$$|\Psi^+\rangle = \frac{|01\rangle + |10\rangle}{\sqrt{2}},$$

and

$$|\Psi^-\rangle = \frac{|01\rangle - |10\rangle}{\sqrt{2}}.$$

For each:

1. Draw the circuit.
2. Display the statevector.
3. Measure 1000 shots.
4. Compare the measurement results.
5. Explain why computational-basis measurement cannot distinguish the + and – relative phases directly.

Exercise 39: Gate matrices using Qiskit

Use Qiskit to obtain the matrix representation of:

- X ,
- H ,
- Z ,
- CNOT.

For example:

```
from qiskit.quantum_info import Operator
```

Compare each Qiskit-generated matrix with the corresponding matrix derived manually.

Exercise 40: Verify matrix multiplication against Qiskit

For the circuit

$$|0\rangle \xrightarrow{H} \xrightarrow{Z} \xrightarrow{H},$$

perform the calculation manually:

$$H Z H |0\rangle .$$

Then implement the same circuit in Qiskit.

Compare:

1. manual matrix multiplication,
2. Qiskit statevector,
3. final measurement result.